Riding the Wave? Tradition and Innovation in Machine Learning Research during the Foundation-Model Shift

Minghan Li
Supervisor: A/Prof Jianzhong Qi

Abstract

Machine learning research moves in waves. At ICLR, the share of submissions building on a foundation model rose from 0.3% in 2017 to 54% in 2026, while the share proposing a new architecture fell from 24% to 8%, and keyword waves such as graph neural networks and federated learning rose and fell within about five years. We ask how a field that changes this fast balances tradition and innovation, at three levels: how the field changes, who brings new work, and what peer review rewards. We study 43,480 ICLR submissions from 2017–2026, including rejected and withdrawn papers, with registered tests run once and replicated on 11,099 held-out papers. Topic maps fitted with hindsight manufacture waves, but displacement is real: where foundation models spread, new architectures fade. Contrary to a herding account, papers by newcomers to a topic more often propose new methods. Review shows no detectable reward for following the crowd; acceptance instead goes with arriving early in a wave and with teams anchored in their own expertise. Innovation comes from outsiders; acceptance favours insiders and early arrivals.

1 Introduction

Machine learning (ML) research changes fast, and it changes in waves. At ICLR, the share of submissions that build on a foundation model rose from 0.3% in 2017 to 54% in 2026, while the share proposing a new architecture fell from 24% to 8%. Author keywords show the same pattern at a finer grain (Figure 1). Generative adversarial networks peaked at 8.1% of submissions in 2018 and have almost disappeared. Graph neural networks, federated learning and self-supervised learning each rose and fell within about five years. Large language models went from nothing to 16.7% of submissions in four years. Even titles follow waves: titles of the form "NAME: subtitle" rose from 9% to 41% of submissions.

Such waves raise an old question about science. Kuhn (1977) called it the *essential tension*: science needs tradition, the shared methods that let work accumulate, and innovation, the departures that open new directions. A fast-moving field makes this tension sharp, in three ways. A wave can displace earlier lines of work (Kuhn, 1962; Bommasani et al., 2021). It can be carried by imitators who follow what is popular (Abrahamson, 1991; Bikhchandani et al., 1992), or by outsiders who bring new ideas (Jeppesen and Lakhani, 2010; Azoulay et al., 2019). And peer review may reward conformity, because novel work is risky and often undervalued (Foster et al., 2015; Wang et al., 2017; Boudreau et al., 2016).

We therefore ask how ML research balances tradition and innovation during the foundation-model shift, at three levels:

RQ1 Field: How does the field change, and can its waves be measured without hindsight?

RQ2 People: Who brings new work, and who follows?

RQ3 Review: Does peer review reward tradition, innovation or following?

We study 43,480 ICLR submissions from 2017 to 2026, built from our own snapshots of official OpenReview records. They include rejected and withdrawn papers, so what we observe is not only what survived review. We place papers in content-based topics using SPECTER2 embeddings (Singh et al., 2023), label each paper's contribution type with a fixed LLM codebook, and identify newcomers to a topic from author identifiers. The core hypotheses were registered before any test was run (Nosek et al., 2018), run once with frozen code, and then tested again on 11,099 held-out papers that we had never analysed. Because we report about twenty findings, each carries an evidence tier,

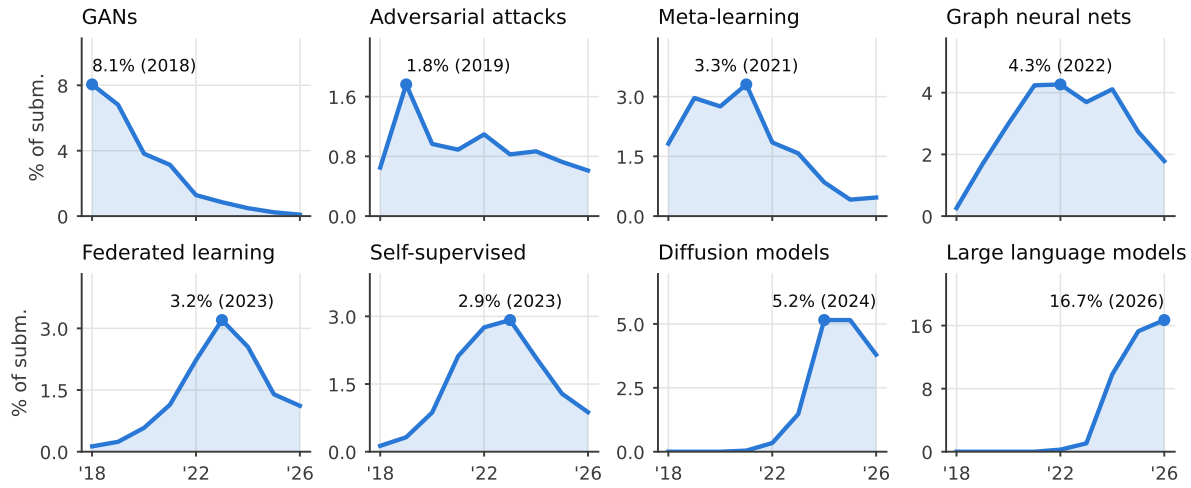


Figure 1: Research waves at ICLR: share of submissions listing each author keyword, 2018–2026 (peak marked). Graph neural networks, federated learning and self-supervised learning are among the 17 of 68 frequent keywords that rose and fell (Section 4).

from "registered and replicated" to "descriptive" (Table 1). All results are associations; none is a causal effect of choosing a topic, a team or a type of contribution.

Field. The waves are real, but the standard way of measuring them is not neutral. A topic map fitted on all years at once, as is common in studies of research trends (Hall et al., 2008), is dominated by recent papers: it finds 29 emerging topics where a map fitted on 2017–2020 papers finds 8. On maps without hindsight, displacement still holds: where foundation models spread, new architectures fade, in discovery and on held-out papers.

People. We registered the herding prediction, and it failed. Papers by newcomers to a topic are more often method papers and less often benchmark papers, in discovery and again on held-out papers. First-time teams are, if anything, slower than other teams to adopt trending keywords and foundation models.

Review. We find no detectable reward for following the crowd: the test for crowded topics is inconclusive, and trendy titles carry no premium. Timing and anchoring matter instead. Papers listing young, rising keywords are accepted slightly more often, and a keyword's acceptance gap falls after its share of submissions peaks; federated learning is the starkest case in discovery. Papers far from their team's own earlier work are accepted less often, and papers whose team includes a topic insider more often.

Together, these findings point to one pattern: *innovation comes from outsiders, while acceptance favours insiders and early arrivals*. Waves go with rewards for those who ride them early, not for those who follow. Our contributions are:

- **Measurement.** We show that topic maps fitted with hindsight manufacture research waves and can reverse trend conclusions, and we give a cheap check: refit the map on early years only.
- **Evidence.** We give a three-level account of how a fast-moving field changes, who drives the change and what review rewards, with every finding labelled by its evidence tier and the core findings replicated on held-out papers.
- **Design.** We show that a registered, outcome-blind, single-run protocol with a held-out replication is feasible for observational studies of science.

2 Background: The Essential Tension

[Part 3.]

3 Data, Measurement and Evidence Tiers

[Part 3.]

4 The Field: Waves, Seen without Hindsight

[Part 2.]

Finding	Discovery	Held-out	Tier
RQ1 Field: how the field changes			
Emerging topics: all-years map vs. 2017–20 map	29 vs. 8	–	E
Keywords that rose and fell within 2018–2026	17 of 68	–	E
Foundation-model (FM) adoption ($0 \rightarrow 100\%$) $\rightarrow$ new-architecture papers	-8.0 [-12.1, -3.8]	-11.0 [-18.1, -3.9]	A
"NAME: subtitle" titles, 2017 $\rightarrow$ 2026	$9\% \rightarrow 41\%$	–	E*
Papers with a same-year twin on the same problem, 2019 $\rightarrow$ 2026	$19.5\% \rightarrow 48.8\%$	–	E
twins per paper per 1,000 submissions	$0.32 \rightarrow 0.19$	–	E
RQ2 People: who brings new work			
Newcomer paper $\rightarrow$ method paper	$+3.7$ [$+2.6$, $+4.8$]	$+3.6$ [$+1.5$, $+5.7$]	A
Newcomer paper $\rightarrow$ benchmark or evaluation paper	-1.7 [-2.5, -0.9]	-2.1 [-3.3, -0.8]	A
Hot topic $\rightarrow$ paper with an incumbent author	-0.7 [-2.7, $+1.2$]	–	C
First-time team $\rightarrow$ FM paper (topics at 5–50% FM adoption)	-4.6 to -14.7	–	E
Returning first authors' lag behind the field: topic vs. title style	$+3\%$ vs. $+40\%$	–	E*
Newcomer paper $\rightarrow$ acceptance; within the same authors	-10.2; -3.1 [-4.5, -1.6]	–	D
RQ3 Review: what is rewarded			
Topic maturity (per log unit of prior papers) $\rightarrow$ acceptance	-1.7 [-3.7, $+0.3$]	-1.5 [-5.2, $+2.2$]	C
"NAME: subtitle" title: share among accepted $\div$ share among all	1.02	–	D*
Young rising vs. old flat keyword $\rightarrow$ acceptance	$+2.7$ [$+0.8$, $+4.6$]	$+3.3$ [-0.2, $+6.7$]	B
Keyword gap after vs. up to its share peak (17 waves)	-5.7 [-8.5, -2.9]	-5.2 [-11.1, $+0.7$]	B
Federated learning keyword $\rightarrow$ acceptance	-11.3 [-17.9, -4.8]	-4.9 [-15.4, $+5.7$]	D
Far jump from the team's own earlier work $\rightarrow$ acceptance	-5.4 [-7.1, -3.8]	-6.1 [-9.3, -2.9]	B
One topic insider vs. none in the team $\rightarrow$ acceptance	$+3.4$ [$+1.2$, $+5.7$]	$+3.9$ [$+0.3$, $+7.5$]	B
FM paper $\rightarrow$ acceptance	$+4.6$	–	D
Topic acceptance edge, 2018–22 $\rightarrow$ 2024–26 (slope)	0.38 [0.18, 0.58]	0.60 [0.32, 0.88]	B

Table 1: All findings, by level. Percentage points (pp) with 95% intervals unless noted; associations within topic and year (keyword rows: against all other papers of the same year). Tiers: **A** registered and replicated on held-out papers; **B** post hoc in discovery, declared before a held-out test that supported it; **C** registered, inconclusive; **D** discovery only (exploratory or post hoc); **E** descriptive, no acceptance outcome. Held-out tests are one-sided, so a supported test can have an interval that includes 0. *Earlier exploratory title analysis; numbers to be re-verified.

5 People: Who Brings New Work

[Part 2.]

6 Review: What Is Rewarded

[Part 2.]

7 Discussion

[Part 3.]

8 Conclusion

[Part 3.]

Limitations

[Part 3 (required by ARR; not counted in the page limit).]

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Draft notes for the supervisor (to be removed)

This draft (Part 1). Title, abstract, introduction and Table 1. The question for this round: does the framing hold? One question (tradition and innovation during the foundation-model shift), three levels (field, people, review) and one through-line (innovation from outsiders; acceptance favours insiders and early arrivals), with every finding placed under a level and labelled by its evidence tier.

Planned figures.

- Figure 1, research waves (this draft).
- Figure 2 (Section 4): topic diversity per year under three topic maps, and emerging topics under each map (hindsight).
- Figure 3 (Section 5): the newcomer coefficients for method and benchmark papers (discovery and held-out), and returning authors' lag behind the field on topic vs. title style.
- Figure 4 (Section 6): federated learning (share of submissions and yearly acceptance gap), next to the wave life cycle (acceptance gap by years since a keyword's peak, pooled over 17 waves).
- Appendix: far jumps from a team's own work, shrinkage of topic acceptance edges, team size, the full held-out table.

Word budget. Abstract 180 (this draft: 179); introduction 650 (this draft: 710); background 350; data 450; field 650; people 550; review 650; discussion 350; conclusion 120. About 3,950 words.

Open questions. (1) Is "tradition and innovation" the right umbrella, or would "conformity and novelty" be clearer? (2) Should federated learning stay a main-text figure, given that it is discovery-only and its held-out test was underpowered? (3) Which rows of Table 1 should move to the appendix?